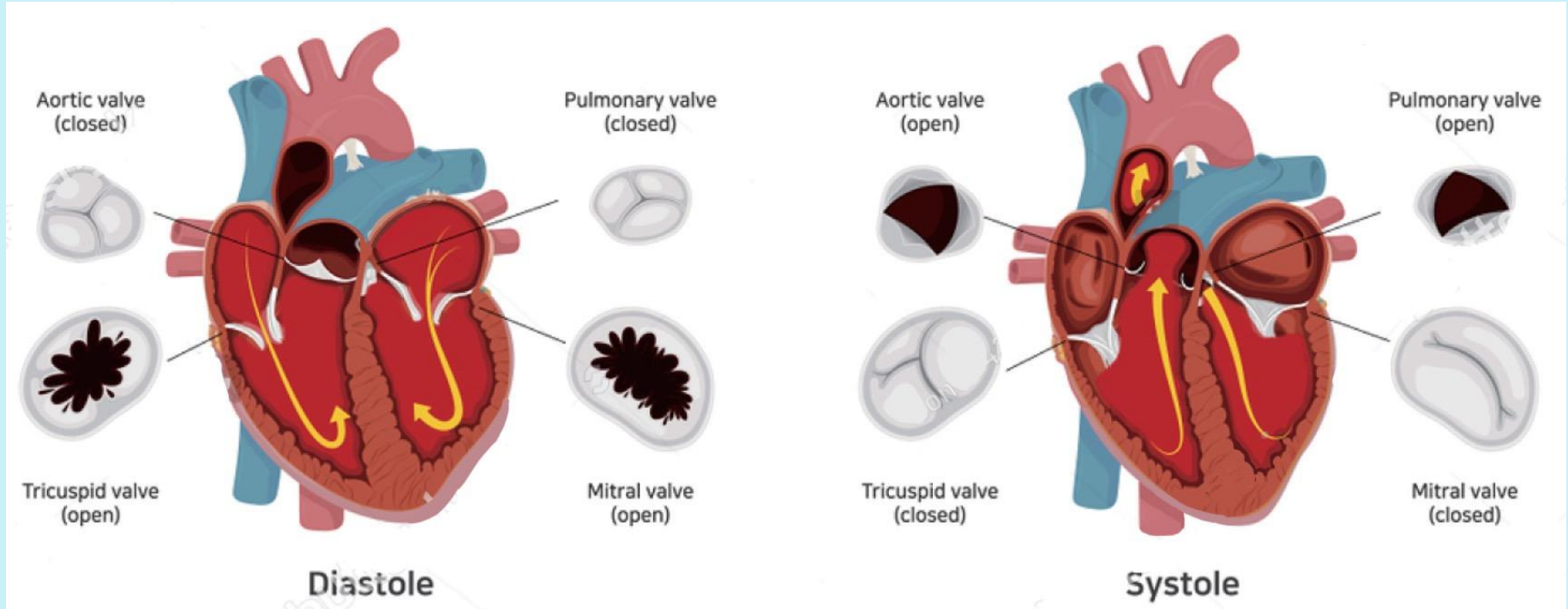


Artificial Heart Valves

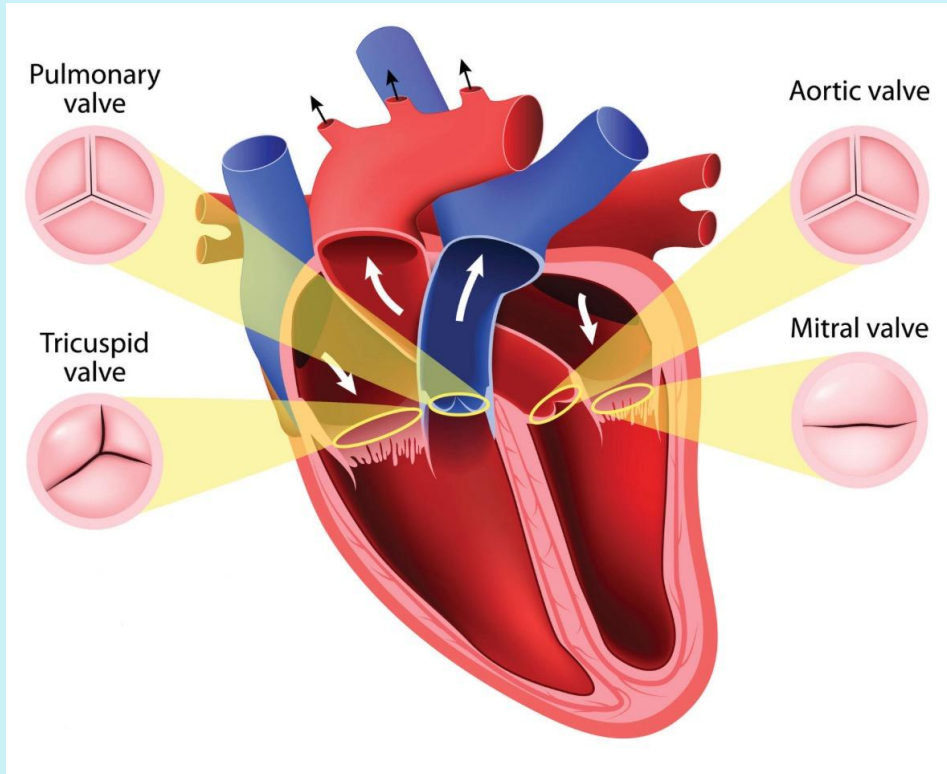
Monday, December 2, 2024

Daniel Hogan, Saba Khan, Moning Pan, Helberth Quysbertf

Heart Valves Function

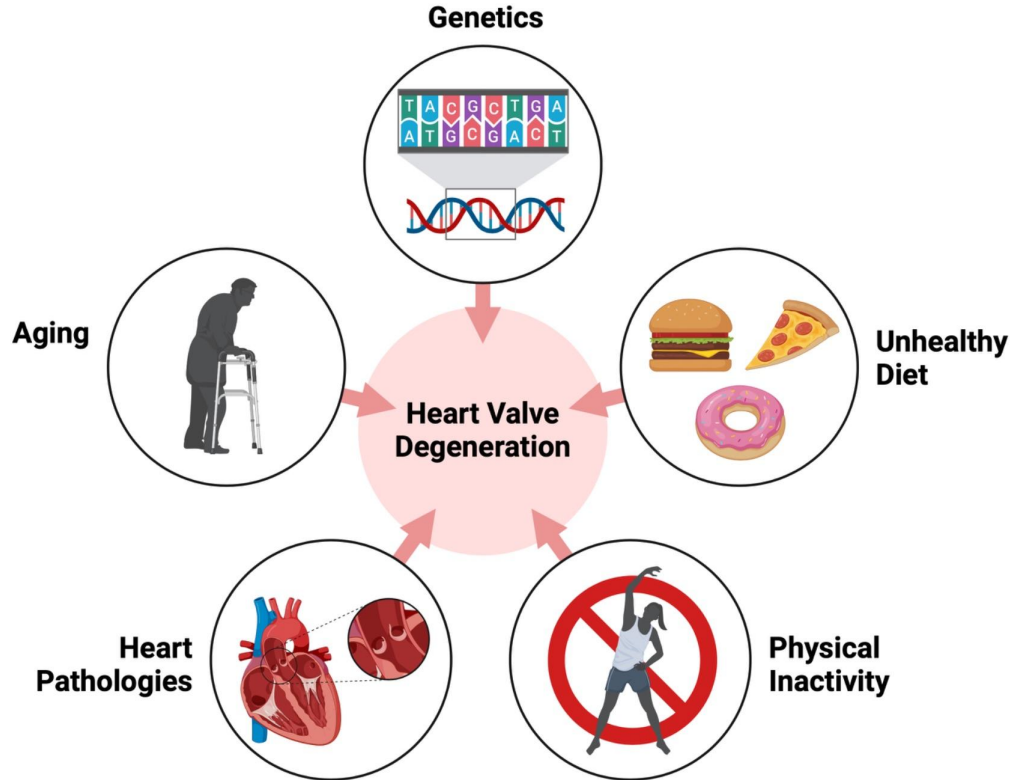


Homeostasis



- **Unidirectionality**
 - Regulate blood flow direction
 - Prevent backflow of blood
- **Overall circulatory health**
 - Maintain cardiac output efficiency
 - Ensure oxygenation and nutrient delivery
- **Must replicate it approximately 3 billion times over an average human lifetime.**

Natural Valves Degeneration



Prosthetic Valves



Mechanical

Advantages:

- Durable, lasts 20+ years
- Resistant to wear and tear
- Suitable for younger patients
- Consistent performance over time

Disadvantages:

- Requires lifelong anticoagulation
- Higher risk of blood clots
- Audible clicking sound possible
- More invasive for future replacements



Biological

Advantages:

- Mimic natural valve function
- Lower risk of blood clots
- No long-term anticoagulation needed
- Quieter operation

Disadvantages:

- Shorter lifespan (10–20 years)
- Prone to calcification over time
- Risk of tissue degeneration
- Limited durability in younger patients

Bi-Leaflet Heart Valve: Market Value

11.1%
INCREASE Projected by 2035

Market Value: \$8.6 billion (2023) → \$30.3 billion (2035)

Key Drivers:

- Rising prevalence of heart diseases and valve replacement procedures.
- Advancements in surgical techniques and prosthetic designs.

Motivations

Solve unmet medical needs

- Reduced dependence on limited biomaterials

Innovate valve technology

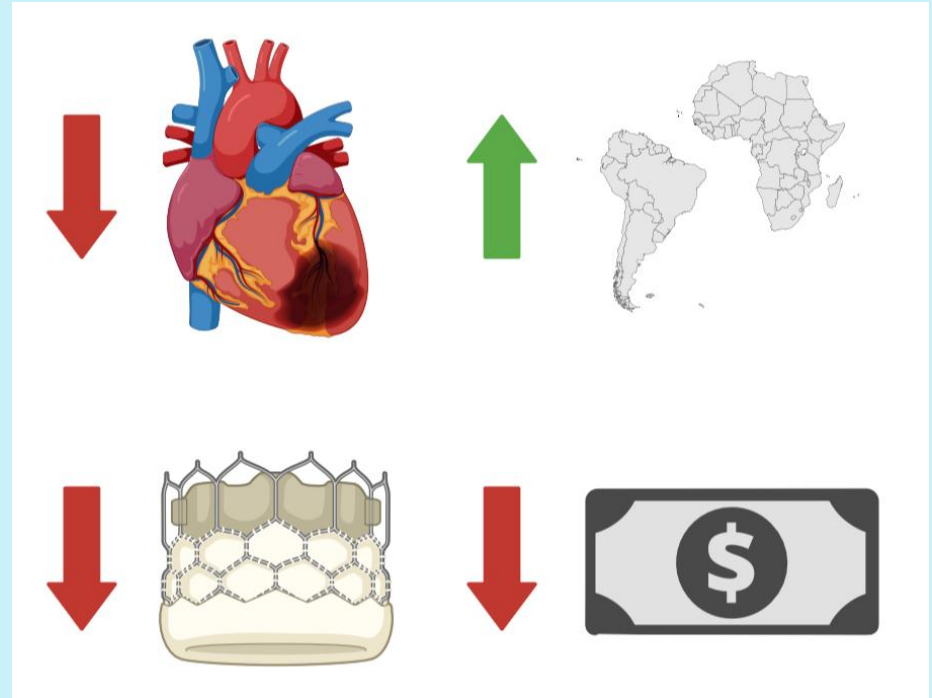
- Reduced thrombogenicity

Address market demand gaps

- Aging population
- Donor compatibility limited

Improve device accessibility

- Reduce the need for repeat surgeries
- Low-income or underserved regions



Past Designs: Caged Ball Valves

Structure & Purpose

Metal cage with a silicone or metal ball.

Cage protected against ball impact; silicone ball reduced wear and enhanced durability

Function

Ball blocked the valve opening to restrict blood flow and moved into the cage to allow flow.

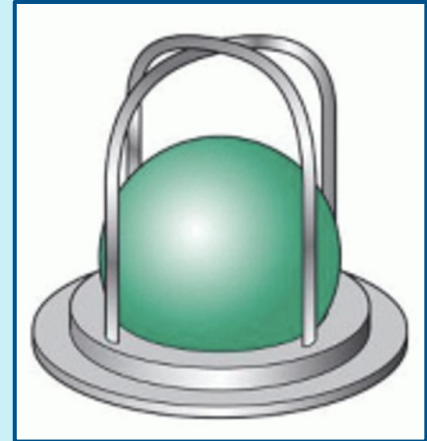
Limitations

Inefficient at mimicking natural blood flow.

Large obstruction caused turbulence and increased pressure gradients.

Blood stagnation increased thromboembolism risk.

Cage structure prone to wear despite valve durability.



Past Designs: Tilting Disk Valves

Structure & Function

Single circular disk pivoting around an axis.

Disk tilts to allow blood flow and closes to prevent backflow.

Advantages

More durable than caged ball valves due to fewer moving parts.

Improved blood flow compared to earlier designs.

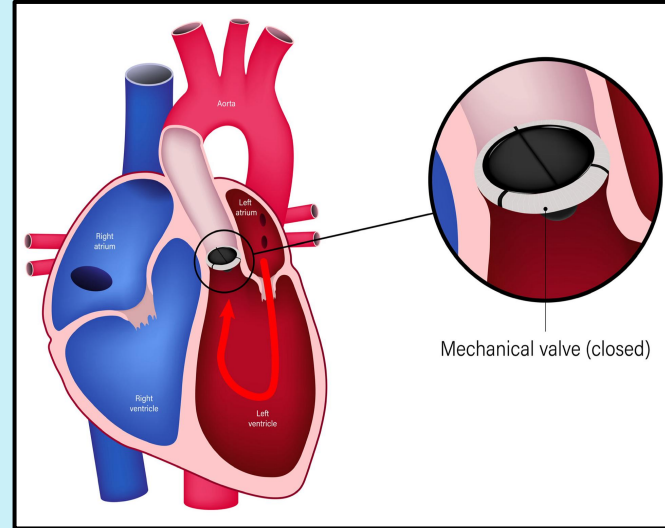
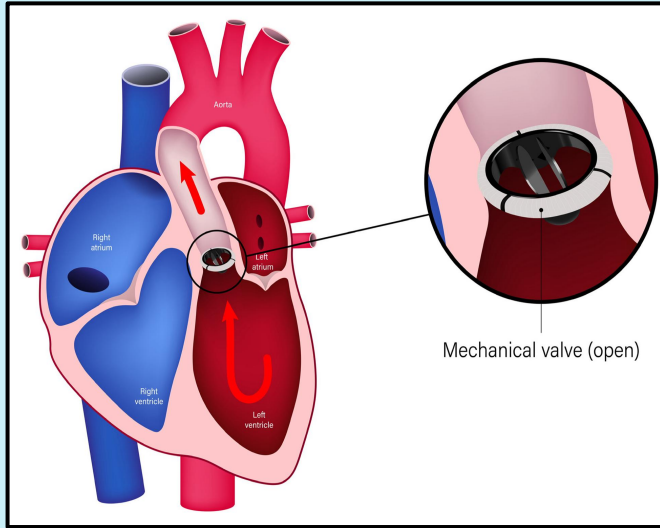
Limitations

Partial obstruction of the orifice.

Increased risk of thromboembolism.



Current Design: Bi-Leaflet Heart Valve



Challenges with Bileaflet Valves

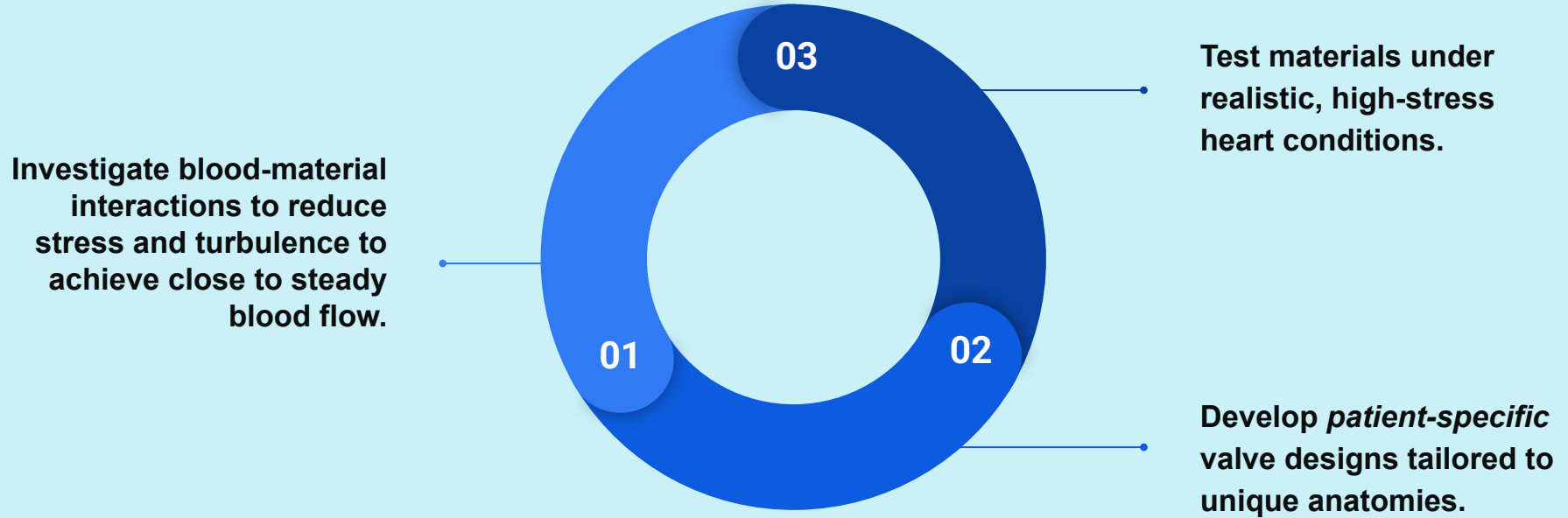
Lifelong Anticoagulation

Persistent risk of thromboembolism due to non-physiological blood flow patterns and turbulence, leading to platelet activation.

Durability Concerns

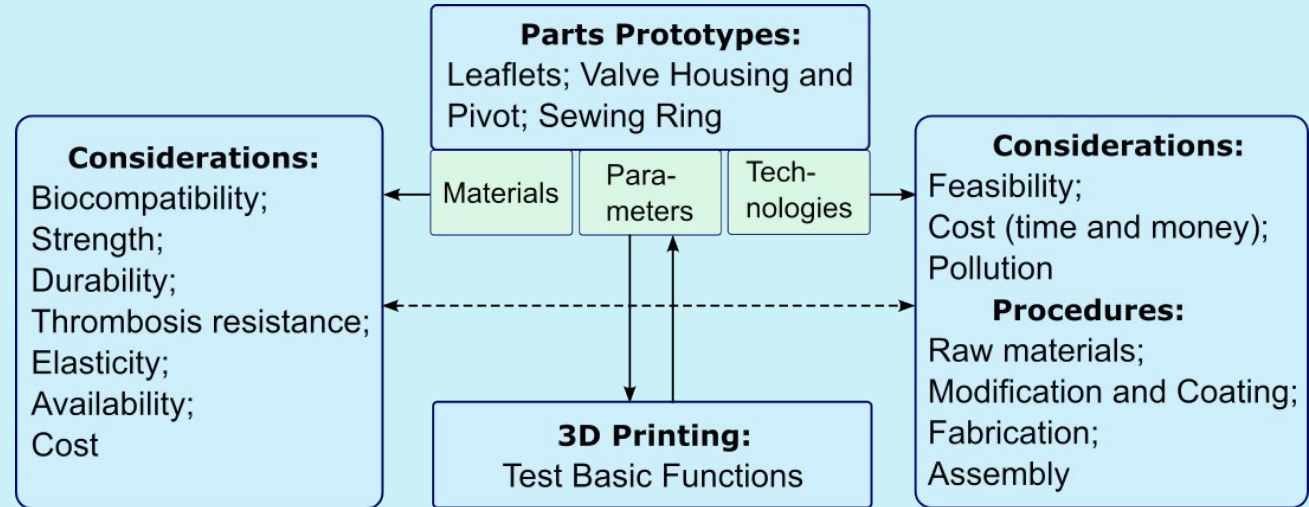
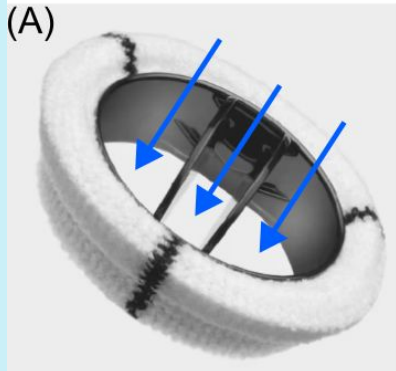
Structural fatigue and wear in leaflet pivots/hinges can cause mechanical failure over time.

Research Needs



Methods

The Roadmap for the design



Methods

Material Selection & Technologies

Leaflets Housing and Pivot Sewing Ring

- Full PyC: Most thromboresistant
- Titanium alloy with PyC coating
- Polytetrafluoroethylene (PTFE) fabric

Surface Modification and Coating Fabrication of Components

- plasma treatment, ion implantation, and UV irradiation; Nanosecond laser etching
- physical vapor deposition (PVD) or chemical vapor deposition (CVD) ; precision laser cutting; thermally process

Results / Discussion

Design Benefits and QFD Chart

Relationships:

Strong	Moderate	Weak	None
9	6	3	0

Customer importance rating	Technical requirements Customer requirements	Hemodynamic Performance	Thrombosis Resistance	Biocompatible Material	Durability	Weighted Score
3	Possible bleeding	6		6	3	45
2	Blood clots	6	9			30
5	Second valve operation	6	6	3	9	120
3	Limitations of daily activities	9	3			36
4	Valve sound	9			3	48
1	Cost	3	9	6	6	24
	Technical importance score	126	66	39	72	

Results / Discussion

Material Benefits & Technical Advantages

Material Benefits

Basic requirements (e.g. biocompatibility, durability and mechanical strength)
Superior requirements (e.g. thromboresistance and flexibility)

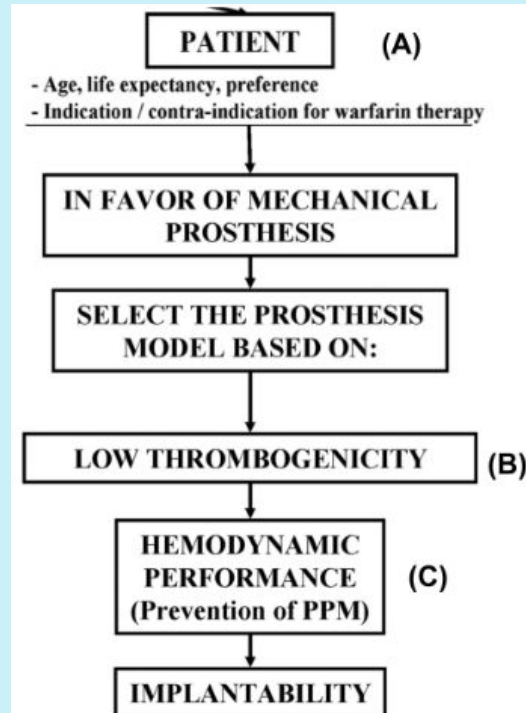
Technical Advantages

Collectively address critical design and functional requirements.
Ensure that the device meets clinical and performance standards.

Critical part requirements	Leaflets	Housing	Sewing ring	Specifications
Dimensional precision	●	● ●	●	±0.01 mm
Surface smoothness	● ●	● ● ●	●	Nano-level
Material strength	● ● ●	● ● ●	●	>100 MPa
Biocompatibility	● ● ●	● ●	●	ISO 10993
Durability	● ● ●	● ● ●	● ● ●	>10 years

Results / Discussion

Summary of Key Results



Algorithm for the selection of the optimal prosthesis in the individual patient.

(A) Initial patient conditions are considered. These conditions include age, life expectancy, preference, and whether or not the patient is relying supplemental therapy

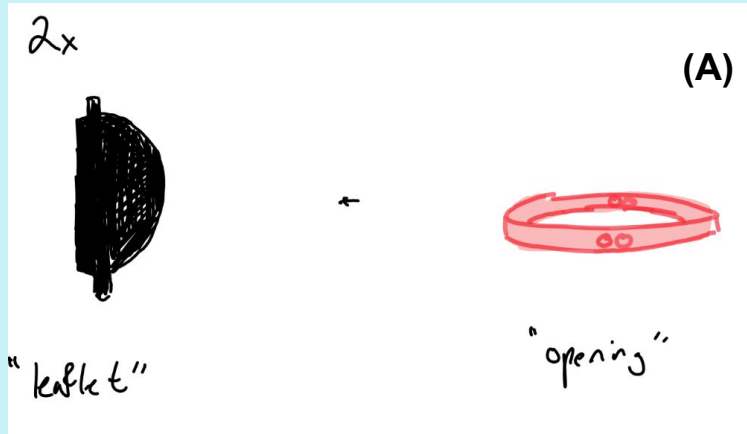
(B) Next, thrombogenicity conditions are considered. Low thrombogenicity is favorable.

(C) Finally hemodynamic performance and implantability are considered. These conditions are patient-dependent and are affected by body size, valve size, age, lifestyle, comorbidities, and left ventricular function.

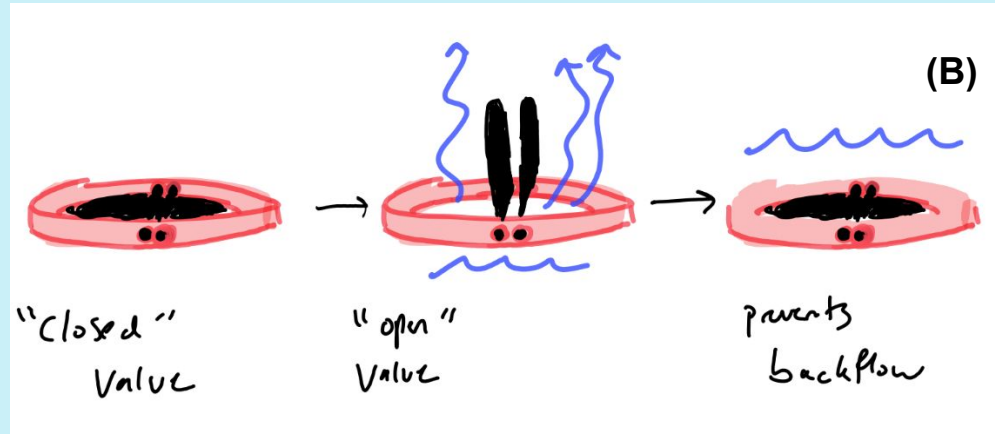
Prototype Design and 3D Printing Trial

Sketches & Early Considerations

Structural Considerations



Functional Considerations



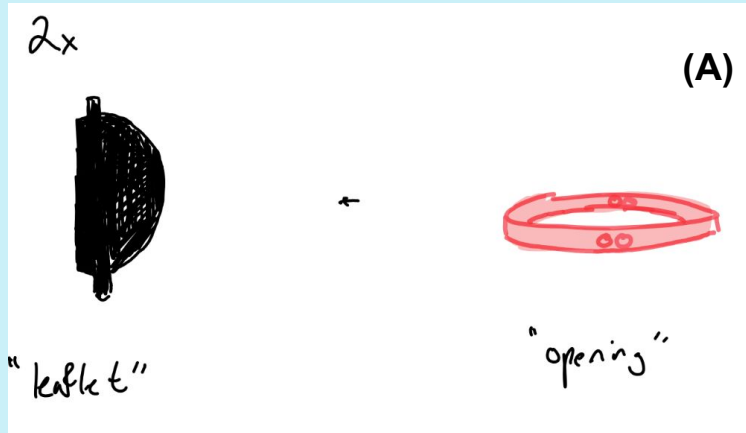
(A) Early sketch featuring the initial structural considerations of the 3D-printed heart valve. On the left shows the "leaflet" part. The right shows the "casing" (in this sketch, called "opening") part.

(B) Early sketch featuring the initial functional considerations of the 3D-printed heart valve. From left-to-right shows the functionality we hoped to achieve, with the valve opening to allow unidirectional flow, but preventing any flow in the opposite direction.

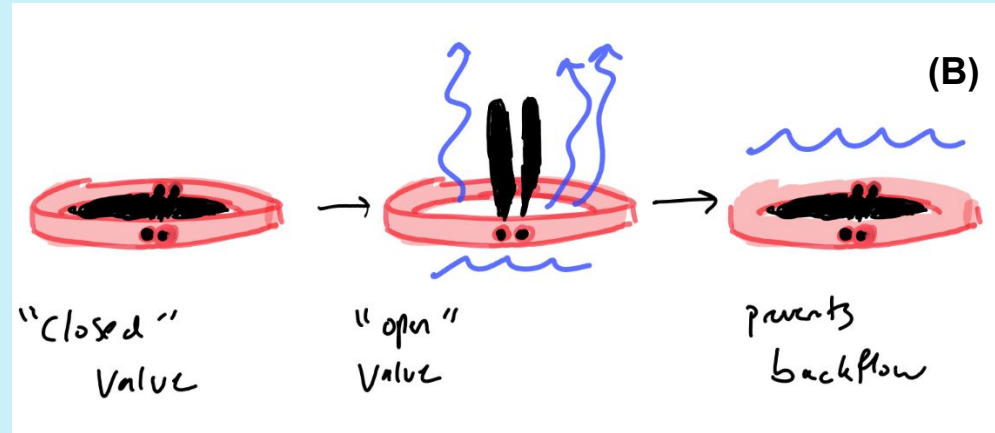
Prototype Design and 3D Printing Trial

Sketches & Early Considerations

Structural Considerations



Functional Considerations



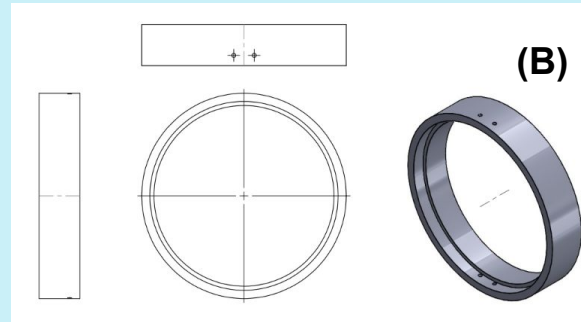
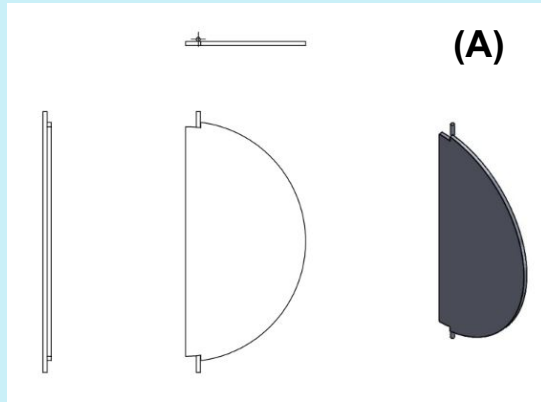
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Prototype Design and 3D Printing Trial

Failed Trials and Revisions

First Printing Trial:



(A) Solidworks design for first printing trial of artificial heart valve leaflet part.

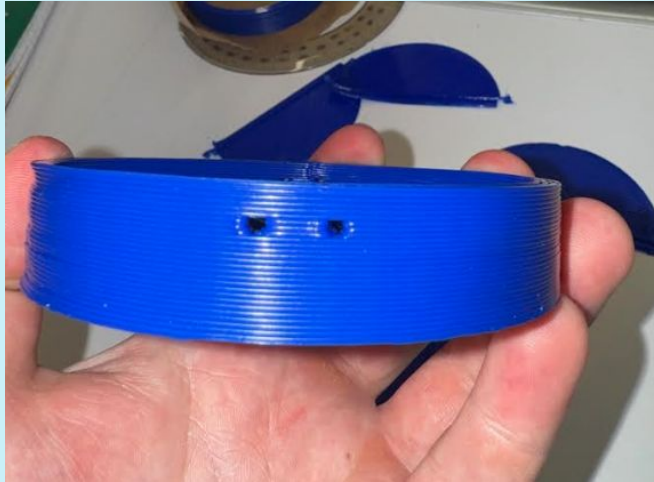
(B) Solidworks design for first printing trial of artificial heart valve casing part.

(C) Two 3D-printed leaflets and casing

Prototype Design and 3D Printing Trial

Failed Trials and Revisions

Errors of the First Design:



Holes on the casing are smaller and more constricted, not allowing for the pegs to be inserted into the casing

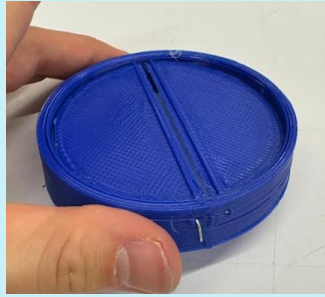
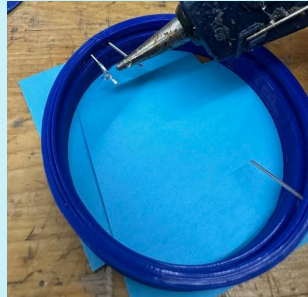
The 3D printing process introduced constraints due to the larger diameter filament used to print. Due to this:

- (1) The pegs on the leaflet were larger than expected. Snapped when trying to be forced into the casing.
- (2) The holes on the casing were constricted

Prototype Design and 3D Printing Trial

Failed Trials and Revisions

Salvaging the First Design:



Metal wire was inserted into the holes of the casing. Hot glue was administered to the metal wires to secure the leaflets. Afterwards, this first design could open in one direction when pressure is applied, but would not open when pressure is applied in the other direction.

Prototype Design and 3D Printing Trial

Failed Trials and Revisions

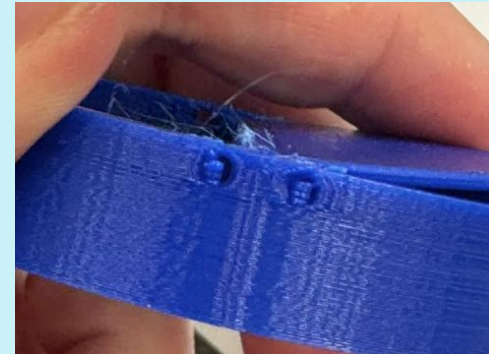
Second Printing Trial: In our second printing trial, we decided to combine both parts into one .stl file and print them both simultaneously.

Pros:

- Avoids any errors associated with popping in the leaflets. (Leaflets will already be in the correct location when the print finishes)
- Quicker print (don't have to do two separate prints)

Cons:

- Leaflet pegs fused with the casing holes during the printing process.
- Caused the leaflets to be stiff and immovable



Fusion of leaflet pegs and casing holes in second printing trial

Prototype Design and 3D Printing Trial

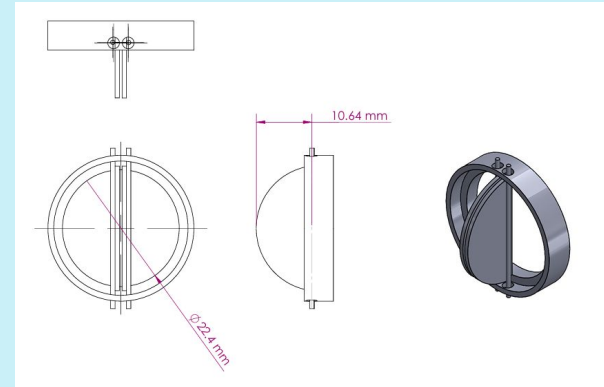
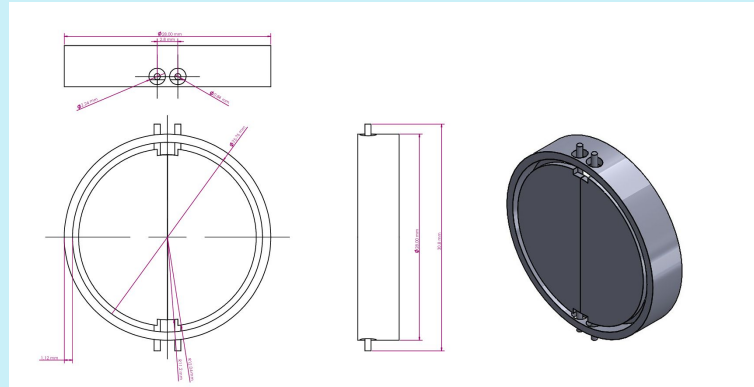
Final Design

Final Design and Printing Trial:

Enlarge the casing holes and made the leaflet pegs longer. Removable scaffolding was printed between the pegs and casing holes.

Pros:

- Leaflets move freely
- No fusion between casing hole and leaflet peg
- All parts printed simultaneously
- Opens in unidirectionally



The final design for the Bileaflet Mechanical Heart Valve.

(A) Solidworks design that depicts the closed position of the valve. Includes measurements.

(B) Solidworks design that depicts the open position of the valve. Includes measurements.

Questions

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